THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 35	$x^3+y^3=(x+y)^3-3xy(x+y)$.
2	**(a)** 98	$x^3-y^3=(x-y)^3+3xy(x-y)$.
3	**(c)** 18	$x^3+\frac{1}{x^3}=\left(x+\frac{1}{x}\right)^3-3\left(x+\frac{1}{x}\right)$.
4	**(b)** 1030301	$101=100+1$; use $(a\pm b)^3$.
5	**(c)** -90	$a^3+b^3+c^3=3abc$ when $a+b+c=0$.
6	**(b)** 31	$2(ab+bc+ca)=(a+b+c)^2-(a^2+b^2+c^2)$.
7	**(d)** 16	Add 2 to both sides.
8	**(d)** $(2x+3)(4x^2-6x+9)$	Use the factorisation identity.
9	**(d)** 133	$x^3+y^3=(x+y)^3-3xy(x+y)$.
10	**(a)** 117	$x^3-y^3=(x-y)^3+3xy(x-y)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=512-288=224$ (1).	1+1
12	$103=100+3$ (1); $103^3=1092727$ (1).	1+1
13	Use $a^3+b^3=(a+b)(a^2-ab+b^2)$; $a=2x$, $b=1$ (1); $=(2x+1)(4x^2-2x+1)$ (1).	1+1
14	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=1000-480=520$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=7^3-3(7)$ (1); $=322$ (1).	1+1+1
16	$a+b+c=0$ ✓ (1); LHS $=1+(64)+(-125)=-60$ (1); RHS $=3(1)(4)(-5)=-60$ (1).	1+1+1
17	Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=2x$, $b=3$ (1); $=(2x-3)(4x^2+6x+9)$ (1).	1+1+1
	OR: $\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=8^3-3(8)$ (1); $=488$ (1).	1+1+1
18	$a+b+c=0$ ✓ (1); LHS $=8+(-343)+(125)=-210$ (1); RHS $=3(2)(-7)(5)=-210$ (1).	1+1+1
	OR: Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=3x$, $b=1$ (1); $=(3x-1)(9x^2+3x+1)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2+y^2=9+56=65$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=27+252$ (1); $x^3-y^3=279$ (1); $(x+y)^2=(x-y)^2+4xy=9+112$ (1); $(x+y)^2=121$ (1).	1+1+1+1+1
	OR: $a^2+b^2+c^2=36-22=14$ (1); $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ (1); Second factor $=14-11=3$ (1); $a^3+b^3+c^3=18+3(6)$ (1); $=36$ (1).	1+1+1+1+1
20	$x^2+y^2=4+48=52$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=8+144$ (1); $x^3-y^3=152$ (1); $(x+y)^2=(x-y)^2+4xy=4+96$ (1); $(x+y)^2=100$ (1).	1+1+1+1+1
	OR: $a^2+b^2+c^2=81-52=29$ (1); $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ (1); Second factor $=29-26=3$ (1); $a^3+b^3+c^3=27+3(24)$ (1); $=99$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 52	$x^3+\frac{1}{x^3}=\left(x+\frac{1}{x}\right)^3-3\left(x+\frac{1}{x}\right)$.
2	**(c)** 970299	$99=100-1$; use $(a\pm b)^3$.
3	**(b)** 36	$a^3+b^3+c^3=3abc$ when $a+b+c=0$.
4	**(b)** 47	$2(ab+bc+ca)=(a+b+c)^2-(a^2+b^2+c^2)$.
5	**(b)** 25	Add 2 to both sides.
6	**(a)** $(3x-2)(9x^2+6x+4)$	Use the factorisation identity.
7	**(b)** 72	$x^3+y^3=(x+y)^3-3xy(x+y)$.
8	**(c)** 37	$x^3-y^3=(x-y)^3+3xy(x-y)$.
9	**(c)** 110	$x^3+\frac{1}{x^3}=\left(x+\frac{1}{x}\right)^3-3\left(x+\frac{1}{x}\right)$.
10	**(d)** 1061208	$102=100+2$; use $(a\pm b)^3$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$97=100-3$ (1); $97^3=912673$ (1).	1+1
12	Use $a^3+b^3=(a+b)(a^2-ab+b^2)$; $a=3x$, $b=2$ (1); $=(3x+2)(9x^2-6x+4)$ (1).	1+1
13	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=216-90=126$ (1).	1+1
14	$105=100+5$ (1); $105^3=1157625$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=9^3-3(9)$ (1); $=702$ (1).	1+1+1
16	$a+b+c=0$ ✓ (1); LHS $=27+(125)+(-512)=-360$ (1); RHS $=3(3)(5)(-8)=-360$ (1).	1+1+1
17	Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=x$, $b=6$ (1); $=(x-6)(x^2+6x+36)$ (1).	1+1+1
	OR: $\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=10^3-3(10)$ (1); $=970$ (1).	1+1+1
18	$a+b+c=0$ ✓ (1); LHS $=-8+(729)+(-343)=378$ (1); RHS $=3(-2)(9)(-7)=378$ (1).	1+1+1
	OR: Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=5x$, $b=2$ (1); $=(5x-2)(25x^2+10x+4)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2+y^2=25+28=53$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=125+210$ (1); $x^3-y^3=335$ (1); $(x+y)^2=(x-y)^2+4xy=25+56$ (1); $(x+y)^2=81$ (1).	1+1+1+1+1
	OR: $a^2+b^2+c^2=144-94=50$ (1); $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ (1); Second factor $=50-47=3$ (1); $a^3+b^3+c^3=36+3(60)$ (1); $=216$ (1).	1+1+1+1+1
20	$x^2+y^2=16+90=106$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=64+540$ (1); $x^3-y^3=604$ (1); $(x+y)^2=(x-y)^2+4xy=16+180$ (1); $(x+y)^2=196$ (1).	1+1+1+1+1
	OR: $a^2+b^2+c^2=100-62=38$ (1); $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ (1); Second factor $=38-31=7$ (1); $a^3+b^3+c^3=70+3(30)$ (1); $=160$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** -210	$a^3+b^3+c^3=3abc$ when $a+b+c=0$.
2	**(b)** 26	$2(ab+bc+ca)=(a+b+c)^2-(a^2+b^2+c^2)$.
3	**(c)** 36	Add 2 to both sides.
4	**(b)** $(x-4)(x^2+4x+16)$	Use the factorisation identity.
5	**(c)** 189	$x^3+y^3=(x+y)^3-3xy(x+y)$.
6	**(c)** 136	$x^3-y^3=(x-y)^3+3xy(x-y)$.
7	**(a)** 198	$x^3+\frac{1}{x^3}=\left(x+\frac{1}{x}\right)^3-3\left(x+\frac{1}{x}\right)$.
8	**(a)** 941192	$98=100-2$; use $(a\pm b)^3$.
9	**(a)** 252	$a^3+b^3+c^3=3abc$ when $a+b+c=0$.
10	**(d)** 74	$2(ab+bc+ca)=(a+b+c)^2-(a^2+b^2+c^2)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use $a^3+b^3=(a+b)(a^2-ab+b^2)$; $a=x$, $b=5$ (1); $=(x+5)(x^2-5x+25)$ (1).	1+1
12	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=1331-792=539$ (1).	1+1
13	$95=100-5$ (1); $95^3=857375$ (1).	1+1
14	Use $a^3+b^3=(a+b)(a^2-ab+b^2)$; $a=4x$, $b=3$ (1); $=(4x+3)(16x^2-12x+9)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=11^3-3(11)$ (1); $=1298$ (1).	1+1+1
16	$a+b+c=0$ ✓ (1); LHS $=216+(1)+(-343)=-126$ (1); RHS $=3(6)(1)(-7)=-126$ (1).	1+1+1
17	Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=4x$, $b=1$ (1); $=(4x-1)(16x^2+4x+1)$ (1).	1+1+1
	OR: $\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=12^3-3(12)$ (1); $=1692$ (1).	1+1+1
18	$a+b+c=0$ ✓ (1); LHS $=64+(64)+(-512)=-384$ (1); RHS $=3(4)(4)(-8)=-384$ (1).	1+1+1
	OR: Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=2x$, $b=5$ (1); $=(2x-5)(4x^2+10x+25)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2+y^2=36+32=68$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=216+288$ (1); $x^3-y^3=504$ (1); $(x+y)^2=(x-y)^2+4xy=36+64$ (1); $(x+y)^2=100$ (1).	1+1+1+1+1
	OR: $a^2+b^2+c^2=49-28=21$ (1); $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ (1); Second factor $=21-14=7$ (1); $a^3+b^3+c^3=49+3(8)$ (1); $=73$ (1).	1+1+1+1+1
20	$x^2+y^2=1+84=85$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=1+126$ (1); $x^3-y^3=127$ (1); $(x+y)^2=(x-y)^2+4xy=1+168$ (1); $(x+y)^2=169$ (1).	1+1+1+1+1
	OR: $a^2+b^2+c^2=225-148=77$ (1); $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ (1); Second factor $=77-74=3$ (1); $a^3+b^3+c^3=45+3(120)$ (1); $=405$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.